



irfu

# PLC PARSE TOOL EPICS Database generator

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IRFU, CEA Saclay, France

EPICS Collaboration Meeting - April 7<sup>th</sup> - 11<sup>th</sup> 2025  
ISIS Neutron and Muon Source



# Technical context

“Limited” skills and available human resource  
The laboratory was divided into 2 groups :

**IT and Automation**

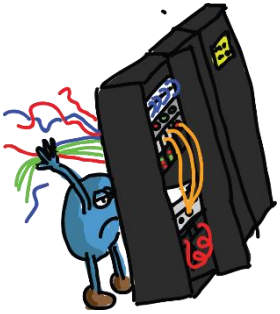


IT developer without expertise in automation



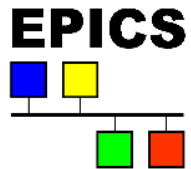
Automation engineer without development skills

# The needs in 1996



Mainly driven by the automation engineer  
EPICS developers focused on ADAS VME Board drivers

**Solution to generate the communication  
between automation and supervision easily**



**Initial release January 19 1999**

Ground test 1988 Bob Dalesio, Jeff Hill, Marty Kraimer  
ICALEPCS 1991, Tsukuba JAPAN hosted by KEK Inter-University  
Accelerators in US : LANL , ANL ...



**Initial released April 19, 2001**

POC demo 2000 Andy Götz, Jean-Michel Chaize, Emmanuel Taurel  
ICALEPCS 1999, Trieste ITALY hosted by Elettra synchrotron  
Synchrotrons in Europe : ESRF, SOLEIL ...





# MUSCADE : $\mu$ Embedded SCADA



- ✓ Embedded in one device (including services)
- ✓ Robust and compact industrial PC (extreme conditions )
- ✓ In-house solution @CEA
- ✓ Only 2 developers
- ✓ Full pure java solution
- ✓ Windows OS
- ✓ SIEMENS PLC S7 drivers
- ✓ Optimized archiving system
- ✓ Advanced alarms management system
- ✓ Secured remote access with certificates



# TOOLKITS TO BUILD SERVER AND SUPERVISION GUI



MUSCADE-AlarmEditor: Folder = D:\www\htm\OPC-UA-PLC1500 [554 Ana / 416 Bool]

File Edit Help

**MUSCADE**

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µSCaDa for Embedded System

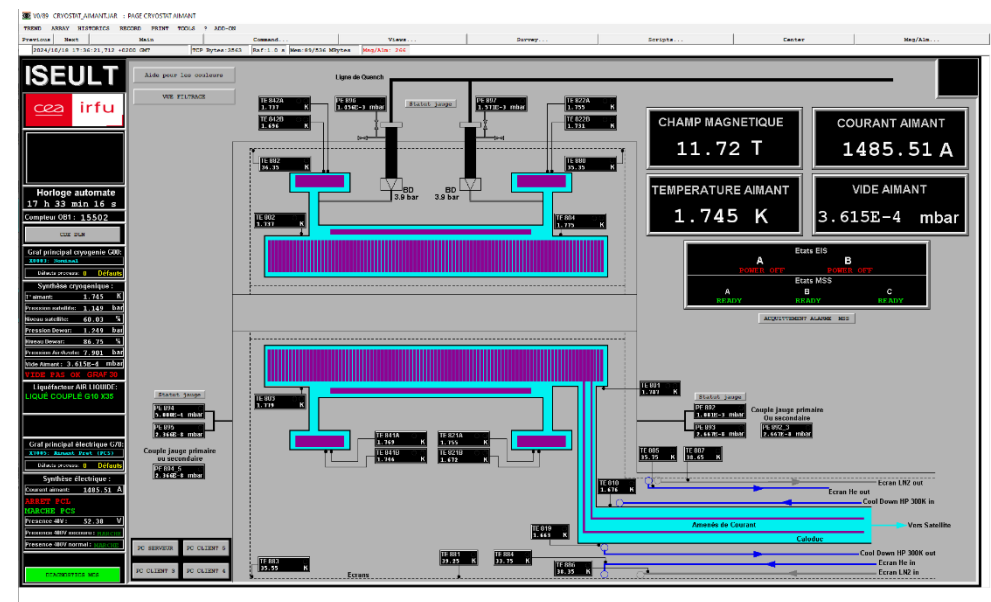
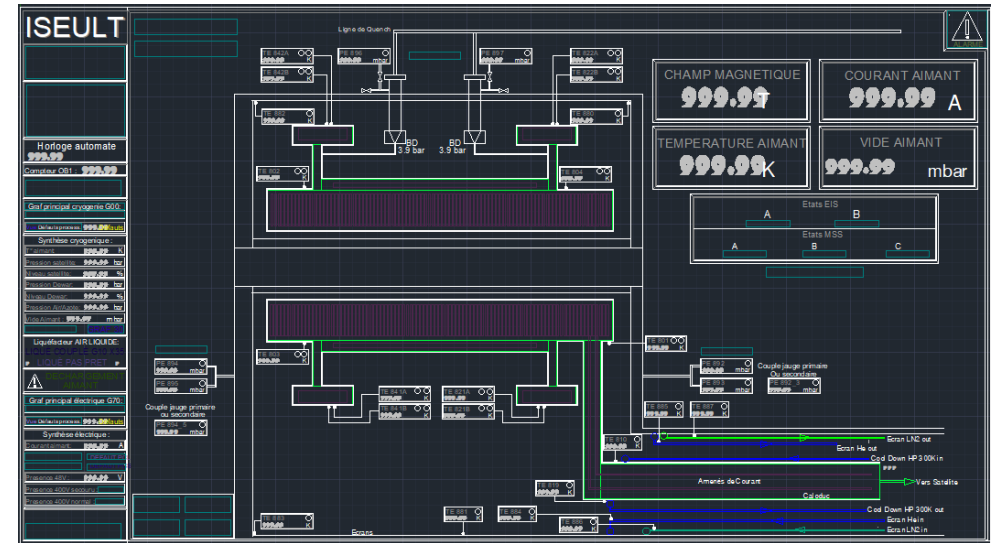
(Compatible with Java SE Oracle JDK 8 / OpenJDK 11)

EMBEDDED SCADA

Analog DataBase (554 items) --- Initial Alarm(s) = 0, Current Alarm(s) = 0

Sort By Active Sort By Name Sort By Comment Sort By List Undo Print Backup... Save

| Num | Active                   | DataName           | Normal Msg | Hyster... | Low Li... | Msg for Low Limit | High Li... | Msg for High Limit | Procedure |
|-----|--------------------------|--------------------|------------|-----------|-----------|-------------------|------------|--------------------|-----------|
| 0   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 1   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 2   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 3   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 4   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 5   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 6   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 7   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 8   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 9   | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 10  | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 11  | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 12  | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 13  | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 14  | <input type="checkbox"/> | PLC_1.ADI_STAAR... |            | 0.5       | -Infinity |                   |            |                    |           |
| 15  | <input type="checkbox"/> | PLC_1.CDE_RWA...   |            | 0.5       | -Infinity |                   |            |                    |           |
| 16  | <input type="checkbox"/> | PLC_1.CDE_RW.C...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 17  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 18  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 19  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 20  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 21  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 22  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 23  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 24  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 25  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 26  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 27  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 28  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 29  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |
| 30  | <input type="checkbox"/> | PLC_1.CDE_RW.D...  |            | 0.5       | -Infinity |                   |            |                    |           |



# MUSCADE KEY FIGURES

Product team : 2022 the 2 experts have retired

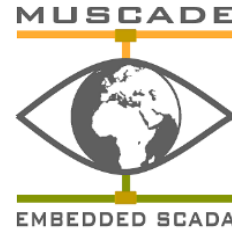
- ✓ 1 developer
- ✓ 3 people in charge of operational support
- Maintenance to keep operational until 2032

Installations :

- ✓ 90 servers
- ✓ 20 projects on CEA site
- ✓ 18 facilities off site
- ✓ 4 facilities supervised with EPICS

Support tracking for 2024 :

- ✓ 185 services tickets opened
- ✓ 62 service requests
- ✓ 17 issues
- ✓ 75 issues on site
- ✓ 46 outside the site (certificate expirations)



## On site

ARIELIX  
ATLAS  
CHYMENE  
CLAS12  
EUCLID  
IPHI\*  
ISEULT  
JT60  
LIQUE 4003  
LIQUE 4008  
LIQUE 4012  
LOTUS  
MAQUETTE  
MGB2  
MINOS  
MIRIM  
SCHEMA  
SOPHI  
STAARQ  
LILIT

## Off site

AGATA\*  
ARTEMIS  
CMS-CDS  
COCOTIER  
COMPASS  
DESI  
ESS\*  
GBAR  
IFMIF\*  
ISEULT  
MEGACAM  
NFS  
R3B  
SARAF  
S3  
SPIRAL2\*  
TAR  
W7X-C1

\* On **EPICS**



**NeuroSpin**  
ISEULT



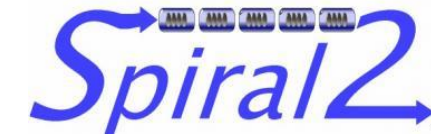
**NOIRLab**  
DESI – Kitt Peak  
Arizona



ArTémis camera APEX  
Chili



1993 TESLA test facility injector - Desy - GERMANY



2011 SPIRAL 2 Heavy-ion sources - Ganil - FRANCE

2013 IFMIF Lipac Accelerator - Rokkasho JAPAN



2014 ESS spallation source – Lund SWEDEN



2017 SARAF Accelerator - Soreq - ISRAEL



2021 AGATA detector - Fermilab - ITALY

2024 PIP II accelerator - Fermilab - ITALY

2025 SEASON detector – Igisol FINLAND



# Mains goals

Provide an intuitive tool to generate COM without requiring IT skills

Reduce human errors and accelerate development

Empower the user with autonomy

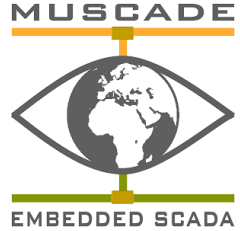
**TO RELY on a TEAM**  
**Eliminate the need for unique expertise**



**“Because alone we go faster but together we go further”**



# SUPPORTED DRIVERS IN BOTH SCADA



**SIEMENS S7**

S7 CEA



✓ Modbus Mark Rivers (Based on Asyn)

✓ S7PLC Dirk Zimoch

✓ S7 CEA support 2020 in house dev CEA

✓ **OPCUA** support 2023 dev ITER Ralph Lange



# STRONG POINTS



**Interoperability** : communication regardless of manufacturer or platform

**Security** : Encryption and authentication

**Data Modeling** : Flexible and with simple addressing via paths “Node Id”

**Standardized Protocol** : Open standard widely adopted and supported



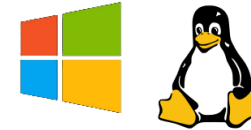
PIP II (Proton Improvement Plan-II) :

- ✓ EPICS OPCUA as main control command system
- ✓ MUSCADE OPCUA for cryogenic cold box system



# PLC PARSER TOOL GENERAL PRINCIPLE

User friendly GUI : Windows and Linux support



1 Protocol selection

2 Siemens TIA Portal file parsing

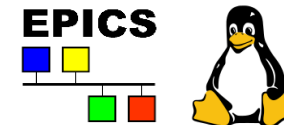
3 Parameters settings

4 Generates EPICS database and protocol configuration

5 EPICS deployment on server



- Done by Automation engineer
- Done by EPICS developer



# PLC PARSETool GUI

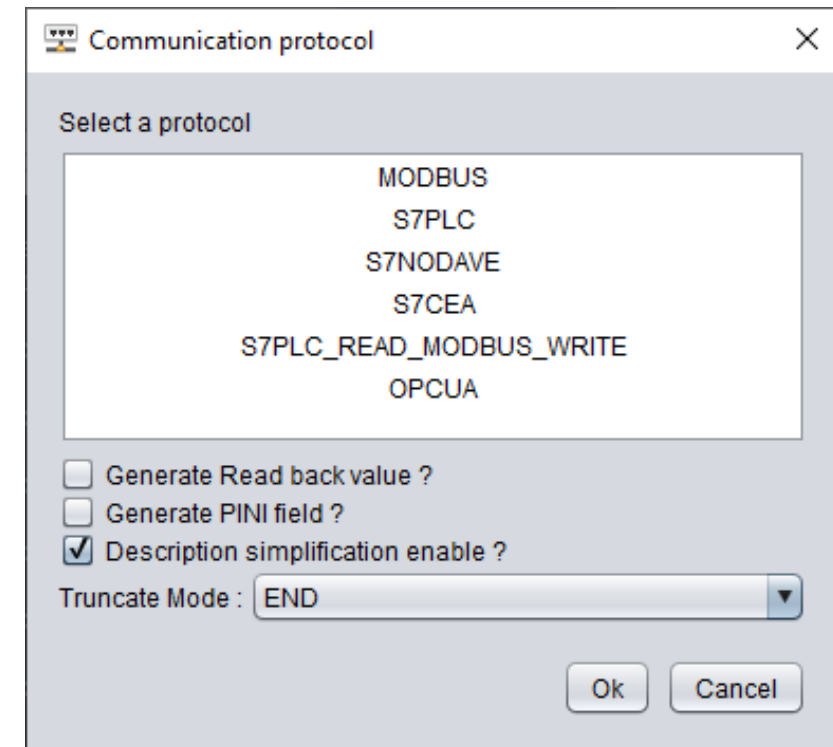
## 1 PROTOCOL SELECTION



Pure Java swing application  
Based on Visual DCT



Protocols options dialog





## 2 OPCUA TIA Portal XML File

~ 7000 variables

```
<?xml version="1.0" encoding="utf-8"?>
<UANodeSet LastModified="2025-02-24T13:08:55Z" xmlns="http://opcfoundation.org/UA/2011/03/UANodeSet.xsd"
  <NamespaceUris>
    <Uri>urn:SIMATIC.S7-1500.OPC-UA.Application:PIP_II_SAF_PLC2</Uri>
    <Uri>http://opcfoundation.org/UA/DI/</Uri>
    <Uri>http://www.siemens.com/simatic-s7-opcua</Uri>
  </NamespaceUris>
  <Extensions>
    <Extension>
      <si:Generator Product="TIA Portal" Edition="STEP 7 Professional" Version="16.0" />
    </Extension>
  </Extensions>
  <UAVariable NodeId="ns=3;s=V_"Controle_Phytron";."Deplacement_D_Grand";" BrowseName="3:Deplace
    <DisplayName>Deplacement_D_Grand</DisplayName>
    <References>
      <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
      <Reference ReferenceType="HasInputVar" IsForward="false">ns=3;s=0_"Controle_Phytron".Inputs</Reference>
    </References>
  </UAVariable>
  <UAVariable NodeId="ns=3;s=V_"Controle_Phytron";."Set Home Position";" BrowseName="3:Set_Home_
    <DisplayName>Set_Home_Position</DisplayName>
    <References>
      <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
      <Reference ReferenceType="HasInputVar" IsForward="false">ns=3;s=0_"Controle_Phytron".Inputs</Reference>
    </References>
  </UAVariable>
  <UAVariable NodeId="ns=3;s=V_"Controle_Phytron";."Cmd_Retour_Home";" BrowseName="3:Cmd_Retour_
    <DisplayName>Cmd_Retour_Home</DisplayName>
    <References>
      <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
      <Reference ReferenceType="HasInputVar" IsForward="false">ns=3;s=0_"Controle_Phytron".Inputs</Reference>
    </References>
  </UAVariable>
```

# 3 PARAMETERS SETTINGS

## Database table based on Visual DCT

| In... | Name                  | Record | Address                               | Data t... | STAT     | DTYP   | TSE | SEVR     | SCAN     |
|-------|-----------------------|--------|---------------------------------------|-----------|----------|--------|-----|----------|----------|
| 1,152 | TELEMETRY_AI_PT218    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AI:IPT218    | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,153 | TELEMETRY_AI_PT110    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AI:IPT110    | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,202 | TELEMETRY_AO_CV410    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV41... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,203 | TELEMETRY_AO_CV450    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV45... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,204 | TELEMETRY_AO_CV465    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV46... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,205 | TELEMETRY_AO_CV469    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV46... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,206 | TELEMETRY_AO_CV470    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV47... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,207 | TELEMETRY_AO_CV481    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV48... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,208 | TELEMETRY_AO_CV482    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV48... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,209 | TELEMETRY_AO_CV483    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV48... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,210 | TELEMETRY_AO_CV9210   | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV92... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,211 | TELEMETRY_AO_CV9110   | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV91... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,212 | TELEMETRY_AO_S_120    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_120   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,213 | TELEMETRY_AO_S_122    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_122   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,214 | TELEMETRY_AO_CV439    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV43... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,215 | TELEMETRY_AO_CV440    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV44... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,216 | TELEMETRY_AO_S_128    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_128   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,217 | TELEMETRY_AO_S_130    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_130   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,218 | TELEMETRY_AO_S_132    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_132   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,219 | TELEMETRY_AO_S_134    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_134   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,220 | TELEMETRY_AO_EH437_SP | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YEH43... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,221 | TELEMETRY_AO_S_138    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_138   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,222 | TELEMETRY_AO_CV490    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV49... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,223 | TELEMETRY_AO_CV493    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YCV49... | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,224 | TELEMETRY_AO_S_144    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_144   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |
| 1,225 | TELEMETRY_AO_S_146    | ai     | @SUB1 ns=3,s=TELEMETRY:I_AOI:YS_146   | REAL      | NO_AL... | OPC... | -2  | NO_ALARM | I/O Intr |

| Date                | Source                      | Type    | Message                                                                                     |
|---------------------|-----------------------------|---------|---------------------------------------------------------------------------------------------|
| 2025/04/04 02:01:20 | PlcParserTool               | INFO    | Configuration file found :C:\Users\KS167484\AppData\Local\Temp\PlcParserTool.config         |
| 2025/04/04 02:01:20 | PlcParserApi                | WARNING | File D:\EPICS\PIPI\AL\Suprathech_Liquéfacteur_S1.MCB.OPCUA.xml.CFG is an unknown plc file   |
| 2025/04/04 02:01:20 | PlcParserApi                | INFO    | Reading file D:\EPICS\PIPI\AL\Suprathech_Liquéfacteur_S1.MCB.OPCUA.xml succeed              |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Update Configuration                                                                        |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Clear records map                                                                           |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Load loc from folder :D:\EPICS\PIPI\AL\locFiles                                             |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Load loc records from db file :D:\EPICS\PIPI\AL\locFiles\locSimatic.s7-1500.opc-uaserver.db |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Load Cmd file D:\EPICS\PIPI\AL\locFiles\locOPCUAConfiguration.cmd                           |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Remove Record                                                                               |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | INFO    | Update Configuration protocol OPCUA                                                         |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | ERROR   | Error on Variable named MemoryClock_0.5Hz contains forbidden character "", rename this var  |
| 2025/04/04 02:01:20 | fr cea ifru epics plcpar... | ERROR   | Error on Variable named MemoryClock_0.625Hz contains forbidden character "", rename this v  |

## EPICS PV Tree

```
graph TD
    Root[Controle_Phytron_1_DB] --> Inputs
    Root --> Outputs
    Root --> Static
    Root --> Timer_Reset_Deplacement
    Root --> Timer_Reset_Retour_Home
    Root --> Controle_Phytron_3_DB
    Root --> Controle_Phytron_4_DB
    Inputs --> Cmd_Retour_Home
    Inputs --> Deplacement_D_Grand
    Inputs --> Deplacement_D_Moyen
    Inputs --> Deplacement_D_Petit
    Inputs --> Deplacement_G_Grand
    Inputs --> Deplacement_G_Moyen
    Inputs --> Deplacement_G_Petit
    Inputs --> Mot_Deplacement_en_cours
    Inputs --> Set_Home_Position
    Outputs --> Static
    Static --> Timer_Reset_Deplacement
    Static --> Timer_Reset_Retour_Home
```

## EPICS & OPCUA parameters

Click here to [Generate IOC Files](#) [EPICS](#)

AppName: SIMATIC.S7-1500.OPC-UA.Application

loc directory path: D:\EPICS\PIPI

PLC name: PIP\_IL\_SAF\_PLC2

PV Prefix:

Ip: 192.168.0.1

Top type: IEE

Description truncate mode: END

Protocol: OPC UA

Port: 4840

OPC session ID: OPC1

Subscription ID: SUB1

Security Mode: Sign

Security Policy: Basic128Rsa15

Update Period ms: 200

Number max of PV for subscription: 20000

Number batch node: 20000

| Date                | Source                      | Type | Message                                                                             |
|---------------------|-----------------------------|------|-------------------------------------------------------------------------------------|
| 2025/04/04 01:43:47 | PlcParserTool               | INFO | Configuration file found :C:\Users\KS167484\AppData\Local\Temp\PlcParserTool.config |
| 2025/04/04 01:43:47 | PlcParserApi                | INFO | Reading file D:\EPICS\PIPI\ISAF_PIP_IL_SAF_PLC2.OPCUA                               |
| 2025/04/04 01:43:47 | fr cea ifru epics plcpar... | INFO | Update Configuration                                                                |
| 2025/04/04 01:43:47 | fr cea ifru epics plcpar... | INFO | Clear records map                                                                   |
| 2025/04/04 01:43:47 | fr cea ifru epics plcpar... | INFO | Load loc from folder :D:\EPICS\PIPI\locFiles                                        |
| 2025/04/04 01:43:47 | fr cea ifru epics plcpar... | INFO | Load loc records from db file :D:\EPICS\PIPI\locFiles\loc                           |

# 4 EPICS DATABASE AND CONFIGURATION GENERATION



```
# OPCUA protocol configuration file
#! Generated by PlcParserTool 2.2.7 the 10/01/2025-19:24:33.958
# add following line in final cmd file
# < iocOpcuaConfiguration.cmd

#=====
#PLC_NAME=MCB

# Server OPCUA address to create a session# opcuaSession sessionId opc.tcp://host:port
opcuaSession OPC1 opc.tcp://test:4840
# Create a subscription on the session define before
# Updating time in ms of this subscription this is usually limit by the server
#opcuaSubscription subscriptionId sessionId updatePeriodInMs
opcuaSubscription SUB1 OPC1 200

# opcuaOptions Switch off/on security
# security mode [None, Sign, SignAndEncrypt]
# opcuaOptions sessionId sec-mode=SecurityMode
opcuaOptions OPC1 sec-mode=None

# Security policy [None, Basic128Rsa15, Basic256, Basic256Sha256,
# Aes128_Sha256_RsaOaep, Aes256_Sha256_RsaPss]
# opcuaOptions sessionId sec-policy=SecurityPolicy
opcuaOptions OPC1 sec-policy=None

# Number of nodes-max = number total of PV for the subscription (max 20000)
# opcuaOptions sessionId nodes-max=nbOfPVMax
opcuaOptions OPC1 nodes-max=20000
# Number of batch-nodes = number total of PV for the subscription
# opcuaOptions sessionId batch-nodes=nbOfBatchNode
opcuaOptions OPC1 batch-nodes=20000

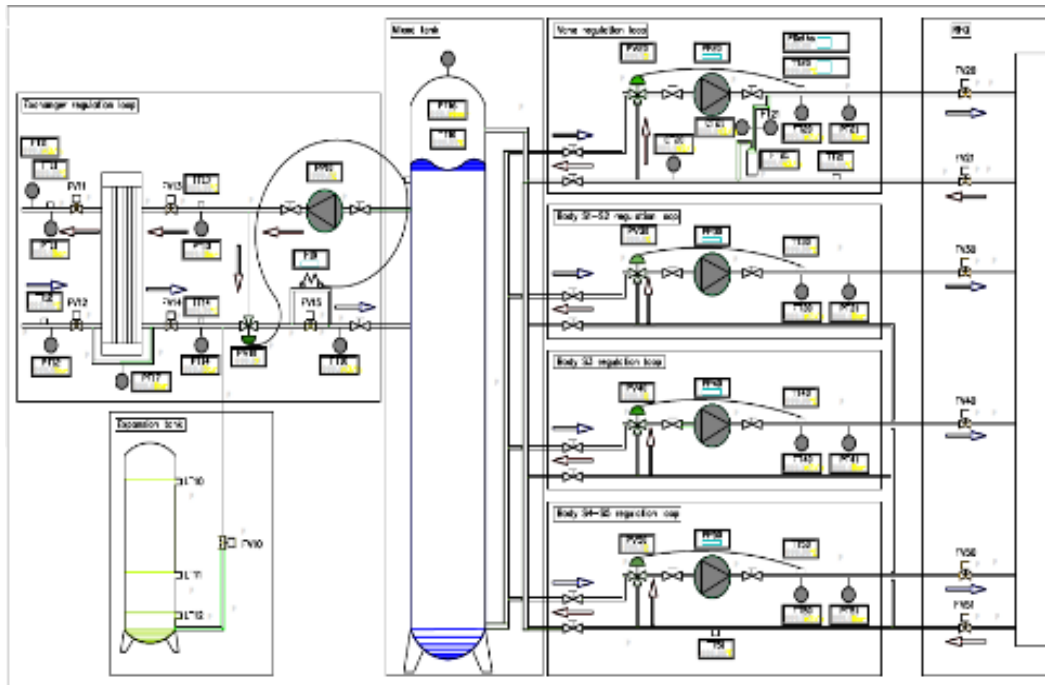
#=====
```



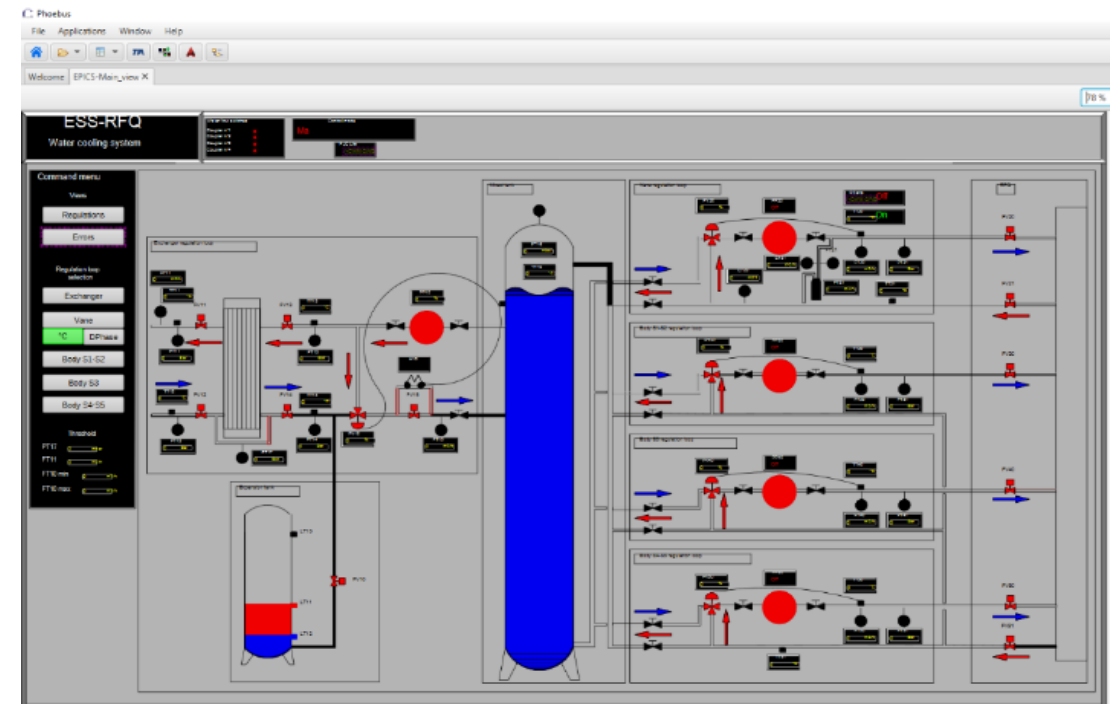
```
D:\EPICS\PIPI\AL\epics\opcua.db - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
opcua.db
14262 field(DTYP, "OPCUA")
14263 field(TSE, "-2")
14264 field(SEVR, "NO_ALARM")
14265 field(INP, "@SUB1 ns=3;s=\"TELEMETRY\".\"_AO\".\"S_138\"")
14266 field(SCAN, "I/O Intr")
14267 }
14268
14269 record(ai, "TELEMETRY:_AO:CV490"){
14270 field(STAT, "NO_ALARM")
14271 field(DTYP, "OPCUA")
14272 field(TSE, "-2")
14273 field(SEVR, "NO_ALARM")
14274 field(INP, "@SUB1 ns=3;s=\"TELEMETRY\".\"_AO\".\"CV490\"")
14275 field(SCAN, "I/O Intr")
14276 }
14277
14278 record(ai, "TELEMETRY:_AO:CV493"){
14279 field(STAT, "NO_ALARM")
14280 field(DTYP, "OPCUA")
14281 field(TSE, "-2")
14282 field(SEVR, "NO_ALARM")
14283 field(INP, "@SUB1 ns=3;s=\"TELEMETRY\".\"_AO\".\"CV493\"")
14284 field(SCAN, "I/O Intr")
14285 }
14286
14287 record(ai, "TELEMETRY:_AO:S_144"){
14288 field(STAT, "NO_ALARM")
14289 field(DTYP, "OPCUA")
14290 field(TSE, "-2")
14291 field(SEVR, "NO_ALARM")
14292 field(INP, "@SUB1 ns=3;s=\"TELEMETRY\".\"_AO\".\"S_144\"")
14293 field(SCAN, "I/O Intr")
14294 }
14295
14296 record(ai, "TELEMETRY:_AO:S_146"){
14297 field(STAT, "NO_ALARM")
14298 field(DTYP, "OPCUA")
14299 field(TSE, "-2")
14300 field(SEVR, "NO_ALARM")
14301 field(INP, "@SUB1 ns=3;s=\"TELEMETRY\".\"_AO\".\"S_146\"")
14302 field(SCAN, ".1 second")
14303 }#! DBDEND
```

# Next steps for GUI developments

- ✓ DXF2BOB and BOB2DXF convertors
- ✓ Presentation in lightning talk “Control System Studio for MUSCADE”



cs|studio







irfu

**Any questions ?**

Thank you for your attention

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Vivien LORIOT  
Paul LOTRUS

EPICS Collaboration Meeting - April 7<sup>th</sup> - 11<sup>th</sup> 2025  
ISIS Neutron and Muon Source

